

User Documentation

**Workshop Management
System**

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1 Overview

The Workshop Management System is a comprehensive digital platform designed to streamline the organization and delivery of educational workshops across various institutions and training providers. This robust system facilitates seamless interaction between three primary user roles: Administrators who oversee the entire system and manage workshops and instructors; Instructors who create and manage sessions, develop tests, and track participant progress; and Participants who can browse workshops, register for sessions, and complete tests for certification. Built with MongoDB and Prisma ORM, the platform offers sophisticated features including intelligent scheduling with conflict detection, integrated payment processing, automated test administration, and certificate generation. The system caters to diverse educational environments, from universities and corporate training departments to independent instructors, providing tools for workshop creation, session management, participant tracking, test administration, and performance monitoring. Through its user-friendly interface and automated processes, the platform significantly reduces administrative overhead while ensuring an engaging and efficient learning experience for all stakeholders.

1.1 System Users

The system caters to three main user groups:

Admin:

- Add workshops and edit them
- Add and edit instructors
- View all the sessions and participants

Trainer:

- View all the workshops, add sessions to the workshop(conflict with the instructor's session and with the sessions of all the workshops with the same venue)
- View all their sessions
- View a session in detail and create tests for the session
- View all the reservations made to their sessions

Member:

- Can view all the workshops and their sessions
- Can register for the sessions(with conflict with their registrations)
- Can register for the tests of the registered sessions
- Can complete the test and view the certificate
- Can view all the reservations

1.2 Key Features and Functionalities

The Workshop Management System incorporates a comprehensive suite of features designed to streamline educational workshop management and enhance the learning experience across all user levels. At its core, the system provides robust administrative controls allowing system administrators to manage workshops, oversee instructor assignments, and monitor overall system activities, while also offering specialized tools for instructors to create and manage sessions, develop assessments, and track participant progress. The platform includes sophisticated scheduling capabilities with built-in conflict detection, ensuring optimal resource utilization across venues and time slots, alongside an integrated payment system for handling workshop fees through multiple payment methods. For instructors, the system offers intuitive tools for session management, test creation, and participant tracking, while participants benefit from easy workshop discovery, seamless registration processes, and access to personal progress tracking and certification. Security is maintained through role-based access control and secure authentication, while the system's automated features handle notifications, scheduling conflicts, and certificate generation. The platform's scalable architecture, built on MongoDB with Prisma ORM, ensures reliable performance and data integrity, making it suitable for educational institutions of any size while maintaining user-friendly interfaces accessible across various devices.

2 Getting Started

2.1 Requirements

To run the application, ensure you have the following software installed:

- **Bun:** v1.0 or latest version
- **Node.js:** 20 or latest version
- **MongoDB:** v4.4 or latest version

2.2 Installation

1. Unzip the project
2. Navigate to the project directory/folder:

```
cd source_code
```

3. Install all the required dependencies/libraries using Bun:

```
bun install
```

3 Configuration

3.1 Environment Variables

The application uses a .env file to manage configuration settings. The following variables should be defined:

- **DATABASE URL:** The URL connection string for MongoDB.
- **SESSION SECRET:** The secret key for session management.

4 Application Structure

The project's framework is Remix-vite and the file structure is as follows:

- **app/:** Contains Remix routes(flat routes), server configuration, api endpoints, mantine components, and tailwind styles.
- **prisma/:** Contains Prisma schema and migration files.

5 Database Management

5.1 Prisma

The application uses Prisma to interact with MongoDB.

1. **Generate Prisma Client:** Run this command after modifying the Prisma schema:

```
bunx prisma generate
```

2. **Seed the database:** Run this command to push the schema to the database and seed it with sample data:

```
bunx prisma db push && bunx prisma db seed
```

6 Running the Application

6.1 Development

To start the application in development mode, run:

```
bun run dev
```

6.2 Production

To build the application for production, run:

```
bun run build
```

To start the application in production mode, run:

```
bun run start
```